

Ἡρώων ὁ Ἀλεξανδρεὺς

Hero of Alexandria on Mechanics, Geometry, and Automata

eul_aid: odi

Period: Ῥωμαϊκὴ [Roman](#)
Dialect: Κοινὴ τεχνικὴ [Technical Koine](#)
Domain: Ἐπιστήμη [Science](#)
Format: Πραγματεία [Treatise](#)

Hero of Alexandria was a Greek mathematician and engineer active in the 1st century CE. His precise dates are unknown, but his work is often placed around 62 CE based on an eclipse mentioned in his **Dioptra**. He worked in Alexandria, and his writings demonstrate extensive knowledge of predecessors like Ctesibius and Archimedes.

His major surviving treatises include **Pneumatica**, describing devices like the steam-powered aeolipile; **Automata**, on mechanical theaters; **Mechanica**, covering simple machines; **Metrica**, which contains Heron's formula for a triangle's area; **Dioptra**, on a surveying instrument; and **Belopoeica**, on catapults. The authorship of **Catoptrica** is sometimes questioned.

Hero is a seminal figure in the history of technology, systematically applying Hellenistic theory to practical engineering. His detailed accounts of complex devices showcase advanced mechanical knowledge, and his mathematical work remained influential for centuries. His writings were preserved in Byzantine and Arabic traditions, impacting later science.

Works (16)

- Βελοποιικά · [Art of War Machines](#) · 35 passages
- Γεωπονικόν · [Book of Agriculture](#) · 171 passages
- Κατοπτρικά · [Catoptrics](#) · 2 passages
- Χειροβαλλίστρας κατασκευή καὶ συμμετρία · [Construction and Symmetry of the Hand Ballista](#) · 22 passages
- Ὅροι · [Definitions](#) · 169 passages
- Διόπτρα · [Dioptra](#) · 47 passages
- Ἀποσπάσματα Μηχανικῶν · [Fragments of Mechanics](#) · 15 passages
- Ἀποσπάσματα περὶ Ὁροσκοπίων · [Fragments-Horoscopes](#) · 41 passages
- Γεωδαισία · [Geodesy](#) · 85 passages
- Γεωμετρικά · [Geometrical Works](#) · 482 passages
- Ἀποσπάσματα Ἡρωνιακά · [Heronian Fragments](#) · 358 passages
- Μετρικά · [Metrical](#) · 95 passages
- Περὶ Αὐτοματοποιητικῶν · [On Automata](#) · 153 passages
- Περὶ Μετρήσεων · [On Measurements](#) · 82 passages
- Πνευματικά · [Spiritual Matters](#) · 93 passages

